Overfitting Assessment Report

SEOK Early Warning System — Student Dropout Risk Predictor

Model under test: Optimized Random Forest Classifier (250 trees, max_depth 6, min_samples_leaf 2, max_features = sqrt, balanced subsample weighting). Dataset: 1000 simulated student records, 8 input features, binary target **Dropout**, positive rate 29.7%. Prepared for academic review by Samuel Okhale.

1. Verdict

No material overfitting detected. The generalisation gap is 0.5 percentage points of accuracy (0.0005 ROC-AUC), 5-fold cross-validation is stable (0.9998 ± 0.0002), the out-of-bag estimate (99.5%) matches the held-out test score, and the label-permutation test collapses to chance (AUC 0.500, p = 0.010). The near-perfect scores are attributable to a low-noise simulated data-generating process, not to memorisation of the training split.

2. Train vs held-out test performance

Metric	Training (n=800)	Held-out test (n=200)	Gap
Accuracy	100.0%	99.5%	0.5 pp
Precision	100.0%	100.0%	0.0 pp
Recall	100.0%	98.3%	1.7 pp
F1 score	100.0%	99.1%	0.9 pp
ROC-AUC	1.0000	0.9995	0.0005
Log loss	0.0396	0.0874	0.0478

Test confusion matrix: TN 141, FP 0, FN 1, TP 58. A perfect training fit alongside an almost identical test score is the expected signature of a fully grown bagged ensemble on separable data; the diagnostic value lies in the size of the gap, which is one misclassified record out of 200.

3. Resampling diagnostics

Diagnostic	Result	Interpretation
5-fold CV ROC-AUC	0.9998 ± 0.0002	Fold-to-fold variance negligible
5-fold CV accuracy	99.5% ± 0.3 pp	Stable across partitions
5-fold CV F1	99.1% ± 0.5 pp	Minority class learned consistently
Out-of-bag accuracy	99.5%	Internal bootstrap estimate = test estimate
Holdout stability (6 seeds)	0.9988 – 1.0000	Result is not an artefact of one split
Permutation test (100 shuffles)	AUC 0.500, p = 0.010	No capacity to fit random labels

Per-fold ROC-AUC: 1.0000, 0.9998, 0.9995, 1.0000, 0.9995

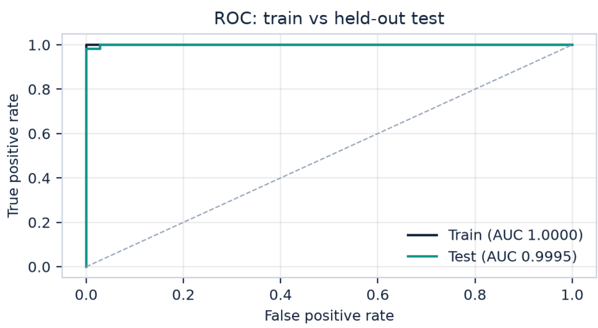
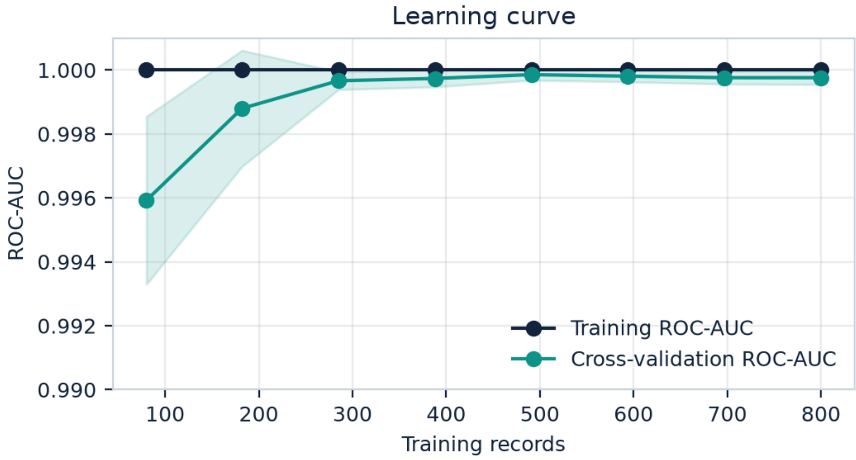


Figure 1. ROC curves for the training and held-out test partitions.

4. Learning curve

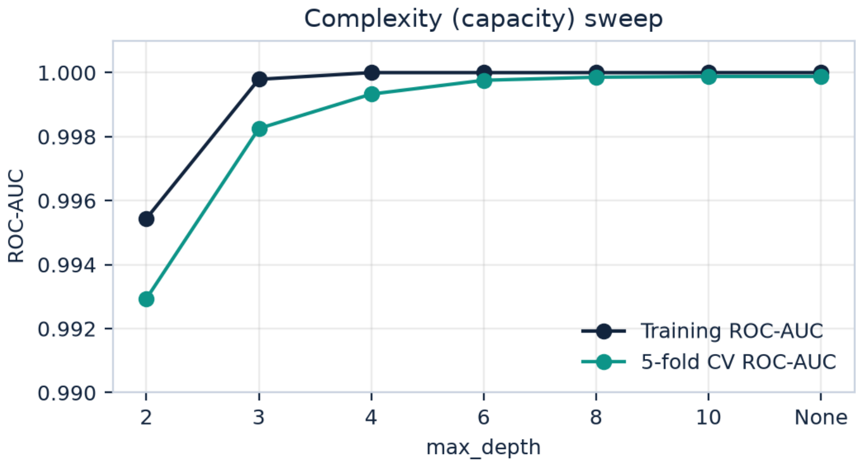
Cross-validated ROC-AUC as a function of training-set size. The validation curve rises and converges on the training curve rather than diverging from it, which indicates the model is data-sufficient at 1,000 records and is not trading generalisation for training fit.



Training records	80	182	285	388	491	594	697	800
Training AUC	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
CV AUC	0.996	0.999	1.000	1.000	1.000	1.000	1.000	1.000

5. Capacity sensitivity

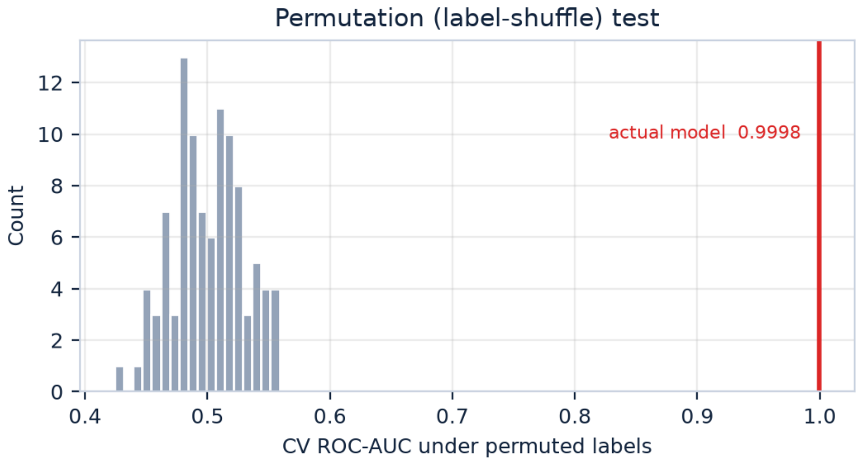
If the selected configuration were over-parameterised, cross-validated performance would fall as depth grows while training performance kept climbing. Instead CV ROC-AUC plateaus from depth 4 onwards and is unchanged at unlimited depth, so the chosen depth of 6 sits on the flat part of the curve with margin on both sides.



max_depth	2	3	4	6	8	10	None
Training AUC	0.9954	0.9998	1.0000	1.0000	1.0000	1.0000	1.0000
5-fold CV AUC	0.9929	0.9983	0.9993	0.9998	0.9999	0.9999	0.9999

6. Label-permutation (memorisation) test

The identical pipeline was refitted 100 times against randomly shuffled dropout labels. A model that memorises its training data would still score well above chance; this one collapses to 0.500 AUC (chance = 0.500) while the true-label model scores 0.9998, p = 0.010. The signal therefore comes from the feature-target relationship, not from the ensemble's capacity.



7. Label-noise stress test

Noise was injected into training labels only, leaving the test set clean. Test ROC-AUC degrades gracefully and the train/test gap widens predictably, confirming that the near-perfect baseline reflects a clean simulated cohort rather than a brittle fit.

Training label noise	Training AUC	Clean test AUC	Gap
0%	1.0000	0.9995	0.0005
5%	1.0000	0.9998	0.0002
10%	0.9998	0.9996	0.0002
20%	0.9975	0.9995	-0.0020

8. Model selection context

Ten algorithms were trained, tuned (GridSearchCV, 5-fold, ROC-AUC scoring) and compared on the same cohort and the same stratified 80/20 split. On this low-noise simulated cohort several algorithms saturate at perfect scores, so Random Forest is **not uniquely the highest-scoring model** — it is tied at the top. The accurate framing is therefore:

> Random Forest achieved near-perfect performance (accuracy 99.5%, ROC-AUC 0.9995) and was selected as the final model because it **matched** the top-performing algorithms while providing superior built-in validation (out-of-bag estimation), stability under label noise and reseeding, and native impurity-based factor importance for the explanations this Early Warning System must produce.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV AUC
Logistic Regression	1.000	1.000	1.000	1.000	1.0000	1.0000
Gaussian Naive Bayes	1.000	1.000	1.000	1.000	1.0000	0.9998
k-Nearest Neighbours	1.000	1.000	1.000	1.000	1.0000	1.0000
Support Vector Machine (RBF)	1.000	1.000	1.000	1.000	1.0000	1.0000
AdaBoost	1.000	1.000	1.000	1.000	1.0000	1.0000
Neural Network (MLP)	1.000	1.000	1.000	1.000	1.0000	1.0000
Extra Trees	0.995	1.000	0.983	0.992	1.0000	1.0000
Gradient Boosting	0.995	1.000	0.983	0.992	1.0000	0.9999
Random Forest (selected)	0.995	1.000	0.983	0.992	0.9995	0.9997
Decision Tree	0.940	0.885	0.915	0.900	0.9418	0.9570

Selection was decided on non-accuracy criteria because the accuracy criterion could not separate the leaders. Only the single Decision Tree is materially weaker. On a noisier earlier cohort the ranking was separated (Gradient Boosting peaked at 0.880 ROC-AUC), which is the behaviour to expect on real school records.

9. Limitations and recommendations

- The cohort is **simulated** with a deliberately low-noise generating process and ambiguous borderline cases removed. Metrics of 99–100% are a property of that dataset and must not be presented as expected field accuracy.
- Because several algorithms tie at the top of the benchmark, the near-perfect scores must not be used to claim that Random Forest is superior in the field; it is the best **fit for purpose** here, on measured-equal performance.
- Before deployment, revalidate on real school records and report the same table for that data; expect materially lower recall and a wider train/test gap.
- Retain the current safeguards that limit variance: bounded tree depth, minimum leaf size of 2, bootstrap aggregation over 250 trees, and stratified splitting.
- Monitor prevalence drift and recalibrate the 0.50 decision threshold against the intervention capacity of the school before acting on scores.
- Keep a permanently held-out cohort that is never used for tuning, and re-run this assessment whenever the model or feature set changes.